

QUESTION PAPER - SET 6

SECTION 1 - PERSONAL DATA

1. Name

2. Mobile Number

3. Class 10 Overall Percentage

4. Class 12 PCM Percentage

5. What is your percentage in Pre Sea Training till 4th Semester?

6. Tell us about your strengths.

7. Tell us about your weaknesses.

8. Have you participated in any extracurricular activities during your Pre Sea Training? If

9. yes, share details.

10. Tell us about your family.

11. Where do you see yourself after 5 years?

SECTION 2 - COMPREHENSION PASSAGE

In the late 21st century, the Global Council enacted the "Resource Equilibrium Accord" to address escalating conflicts over dwindling freshwater reserves. The Accord introduced a tiered quota system, assigning water access rights based on each nation's ecological footprint, technological capability for desalination, and per-capita consumption history.

While wealthier nations invested in advanced desalination plants and atmospheric water harvesters, poorer nations — despite having smaller ecological footprints — struggled to meet their citizens' basic needs under their reduced quotas.

Critics argued that the Accord unintentionally entrenched inequality, as historical consumption patterns favored nations that had previously overused water.

To address the backlash, an amendment proposed the creation of a "Water Equity Fund" financed by a surcharge on nations exceeding their quota. The fund would support sustainable water projects in under-resourced regions.

Yet, debates continued over enforcement: how to monitor usage accurately, prevent data manipulation, and ensure that the fund genuinely benefited those in need.

1. What was the main goal behind the introduction of the Resource Equilibrium Accord in the late 21st century?

- A. To establish a global water trading market
- B. To balance global water distribution and prevent conflicts over freshwater resources
- C. To reward nations with low per-capita water consumption
- D. To increase funding for desalination research

2. Which aspect most contributed to the Resource Equilibrium Accord unintentionally reinforcing inequality among nations?

- A. Technological advancements in desalination
- B. Inaccurate monitoring of water usage
- C. Historical consumption patterns that benefited countries with prior high water use
- D. Opposition to the Water Equity Fund

3. Which conclusion is best supported by the information provided in the passage about the Resource Equilibrium Accord?

- A. The amendment eliminated all concerns about fairness
- B. Nations with smaller ecological footprints were given larger quotas
- C. Wealthier nations opposed atmospheric water harvesting
- D. The Accord unintentionally reinforced pre-existing inequalities

4. Under the Resource Equilibrium Accord, when would a nation be required to pay a surcharge to support the Water Equity Fund?

- A. When its per-capita consumption is below the global average

- B. When it fails to reduce its ecological footprint
- C. When it does not participate in atmospheric water harvesting
- D. When its water usage exceeds the assigned quota

5. What is the main challenge that remains unresolved regarding the amendment to the Resource Equilibrium Accord?

- A. Lack of sustainable water project ideas
- B. The unwillingness of wealthier nations to pay surcharges
- C. Uncertainty in the fund's revenue source
- D. Ensuring transparency and preventing misuse of funds

SECTION 3 - COMPREHENSION PASSAGE

The Sundarbans, located in the delta region of India and Bangladesh, is the largest mangrove forest in the world. It is home to diverse wildlife, including the famous Royal Bengal Tiger. The mangroves act as a natural barrier, protecting coastal areas from cyclones and storm surges. They also provide livelihoods for local communities through fishing, honey collection, and eco-tourism. However, climate change poses a serious threat to the Sundarbans. Rising sea levels, increasing salinity, and frequent storms are causing habitat loss.

Conservation projects are being implemented, focusing on planting more mangroves, reducing illegal logging, and promoting sustainable fishing practices. International cooperation is considered vital for the long-term preservation of this unique ecosystem.

1. Which of the following is NOT listed as a livelihood activity for local communities in the Sundarbans, according to the passage? (1 Point) Select the option that is not mentioned as a source of livelihood.

- A. Agriculture
- B. Honey collection
- C. Fishing
- D. Eco-tourism

2. Which of the following is a direct impact of climate change on the Sundarbans ecosystem? (1 Point) Select the option that best represents a climate change effect described in the passage.

- A. Increasing salinity of water
- B. Introduction of new wildlife species
- C. Growth in honey collection
- D. Expansion of fishing zones

3. What is one of the key protective roles played by the mangrove forests in the Sundarbans? (1 Point) Select the option that best describes how mangroves benefit the coastal environment.

- A. They increase the frequency of tidal waves.
- B. They are a major source of underground minerals.
- C. They supply drinking water to nearby villages.
- D. They act as a shield against cyclones and storm surges.

4. What is the primary objective of the conservation efforts described in the Sundarbans passage? (1 Point) Select the option that best reflects the main goal of these projects.

- A. To increase honey production for export
- B. To encourage local communities to move to urban areas
- C. To restore and safeguard the mangrove forests and their biodiversity
- D. To expand commercial fishing operations

5. Why does the passage emphasize the necessity of international cooperation for preserving the Sundarbans? (1 Point) Select the option that best explains the importance of cross-border collaboration.

- A. The Sundarbans ecosystem spans across India and Bangladesh, requiring joint efforts.
- B. Local communities are unwilling to support conservation projects.
- C. The region attracts international tourists who fund conservation.
- D. Climate change is only a concern for a few developed countries.

SECTION 4 - ENGLISH APTITUDE

1. What is Software Testing?

- a) Analyzing documentation
- b) Writing code
- c) Executing a system to identify bugs and errors
- d) Compiling source code

2. What is a test case?

- a) A list of all the bugs in the software
- b) A document that describes how a test should be executed
- c) A report of testing results
- d) The final version of the software

3. Which of the following is NOT a principle of software testing?

- a) Testing shows the presence of defects
- b) Testing is best done only at the end of development
- c) Early testing saves time and cost
- d) Exhaustive testing is impossible

SECTION 5 - NUMERICAL APTITUDE

1. What is Regression Testing?

- a) Testing only new features
- b) Conducting user interface checks
- c) Validating database constraints
- d) Re-testing existing code to ensure new changes don't affect functionality

2. What is User Acceptance Testing (UAT)?

- a) Debugging system errors
- b) Running automated test execution
- c) Validating business requirements with end-users
- d) Testing system architecture

3. What is the role of a test manager?

- a) To write code for the software
- b) To deploy the software to production
- c) To develop the test environment
- d) To oversee the testing process and ensure test cases are executed

SECTION 6 - MARITIME KNOWLEDGE

1. In radar navigation, what is the main purpose of using a parallel index line?

- A. Determining gyro error
- B. Maintaining a constant distance from hazards or the coastline
- C. Plotting tidal set and drift
- D. Calculating the vessel's speed

2. Which international organization regulates global shipping standards?

- A. NASA
- B. WHO
- C. IMO
- D. UNESCO

3. Which instrument is specifically used to measure the apparent wind direction and speed on a moving ship?

- A. Barometer
- B. Psychrometer
- C. Hygrometer
- D. Anemometer

4. A navigational aid emits light for a shorter time than it remains dark in each cycle. What is this light called?

- A. Isophase
- B. Occulting
- C. Intermittent
- D. Flashing

5. In navigation, what does “dead reckoning” primarily rely upon?

- A. Radar observations

- B. Celestial sightings
- C. Estimated course and speed
- D. Satellite communication

6. What is the primary navigational purpose of observing the Sun at its meridian passage?

- A. Applying time correction
- B. Finding longitude
- C. Calculating compass error
- D. Determining latitude

7. Which method is most commonly used to determine compass error during sunrise or sunset?

- A. Latitude by meridian passage
- B. Azimuth
- C. Sextant altitude
- D. Amplitude

8. At which location on Earth does the Coriolis effect reach its greatest strength?

- A. At the Equator
- B. At the Tropics
- C. At 30° latitude
- D. At the Poles

9. In the Northern Hemisphere, what is the typical direction of wind circulation around a low-pressure area?

- A. Anticlockwise and inward
- B. Clockwise and inward
- C. Clockwise and outward
- D. Anticlockwise and outward

10. Which of the following corrections must be applied to a sextant reading to accurately determine the observed altitude (Ho) of a celestial body?

- A. Index error, dip, refraction, semi-diameter & parallax
- B. Index error and dip only
- C. Refraction only
- D. Dip and refraction only

11. In celestial navigation, what is the First Point of Aries () primarily used to measure?

- A. Measuring True Altitude
- B. Measuring Azimuth
- C. Measuring Right Ascension
- D. Measuring Longitude

12. What does the term "angle of loll" describe in ship stability?

- A. The angle at which the center of gravity is vertically above the center of buoyancy
- B. The time taken for a vessel to complete one roll cycle
- C. KM is greater than GM
- D. The heel angle at which a vessel with negative GM achieves equilibrium due to hull shape

13. When crossing a traffic separation scheme, the vessel should:

- A. Cross at any convenient angle
- B. Cross at right angles to the traffic flow
- C. Alter course to avoid lanes
- D. Follow the traffic flow

14. Which cloud type is most commonly associated with thunderstorms and heavy rainfall?

- A. Cirrus
- B. Stratus

- C. Cumulonimbus
- D. Altocumulus

15. What is the purpose of a ship's whistle or horn?

- A. Entertainment
- B. Distress communication only
- C. Navigational sound signaling
- D. Engine warning

16. Which instrument is used to measure the salinity of seawater?

- A. Hygrometer
- B. Sextant
- C. Salinometer
- D. Barograph

17. Which factor most affects the rolling period of a ship?

- A. Cargo color
- B. Metacentric height
- C. Number of lifeboats
- D. Radar frequency

18. The total area beneath the GZ (righting arm) curve from 0° to 40° of heel is a measure of which of the following vessel characteristics?

- A. The vessel's dynamic stability within that heel range
- B. The vessel's metacentric height (GM)
- C. The vessel's KM
- D. The vessel's maximum righting lever

19. The azimuth of a celestial body is measured at which of the following locations?

- A. Observer's Zenith
- B. Observer's Nadir
- C. Observer's Meridian

D. Observer's Prime Vertical

20. Which opening below is NOT considered while assessing Conditions of Assignment?

- A. Hatchway on freeboard deck
- B. Door in superstructure end bulkhead
- C. Sounding pipe opening in engine room
- D. Ventilator on freeboard deck

21. Which document contains detailed information about cargo carried on a vessel?

- A. Deck Log Book
- B. Cargo Manifest
- C. Engine Log Book
- D. Stability Booklet

22. A vessel alters course from 090° to 180°. What is the amount of alteration?

- A. 45°
- B. 90°
- C. 180°
- D. 270°

23. If 1800 metric tonnes of cargo are loaded, what will be the vessel's new mean draft?

- A. 10.10 m
- B. 9.19 m
- C. 9.80 m
- D. 9.99 m

24. The observer's prime vertical passes through the east and west points of which of the following?

- A. Rational horizon

- B. Equinoctial
- C. Celestial meridian
- D. Great circle

25. How should a newly issued Preliminary Notice to Mariners (P-Notice) be marked on a nautical chart?

- A. In red marker
- B. Lightly in pencil
- C. In permanent ink
- D. Not marked until confirmed

26. What is the main purpose of a ship's anchor?

- A. To increase vessel speed
- B. To hold the vessel in position
- C. To improve fuel efficiency
- D. To reduce rolling motion

27. In ship stability, GM refers to:

- A. Gross Measurement
- B. General Metric
- C. Metacentric Height
- D. Gravity Measurement

28. Aneroid barometers are typically calibrated to display atmospheric pressure in which of the following units?

- A. Inches of mercury and millibars
- B. Feet of mercury and millibars
- C. Inches of mercury and millimeters
- D. Inches of mercury and centimeters

29. What does "port side" mean on a vessel?

- A. Right side
- B. Left side

- C. Front side
- D. Upper deck

30. What is the primary purpose of a ship's radar?

- A. Cooking assistance
- B. Detecting objects and navigation
- C. Measuring cargo weight
- D. Monitoring fuel usage